

University of Waterloo
Department of Systems Design Engineering
Biomedical Engineering Program

Design Showcase 1
Therapeutic Assistive Devices in Aquatic
Environments for Children with Cerebral Palsy

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1. Needs Assessment

1.1 Situation of Concern

To develop an effective and realistic situation of concern, we started by researching the physical abilities of toddlers with all types of cerebral palsy (spastic, dyskinetic, ataxic and mixed) [1], this acted as the foundation for our problem space while we conducted following research for the situation of concern.

While there were many potential stakeholders, the stakeholders we could identify for certain were the child with cerebral palsy as well as their parents or primary caretakers. Other stakeholders such as pediatricians, occupational therapists, and support systems (such as other families of children with cerebral palsy) could vary from case to case but could be assumed for most cases. We also thought about other possible support groups, such as teachers and babysitters, however the existence of these stakeholders would require further research into the specific child using the solution. Understanding the potential stakeholders involved allowed for identification of a constraint of our design: it needed to be accessible and easy for so many types of potential users to understand and maintain.

To identify the processes required by the primary user, we conducted research into what it means to have level 4 to 5 cerebral palsy. According to our research, children with severe cerebral palsy likely have trouble performing most physical movements, especially those requiring fine motor skills, making this a process our design needed to address [2]. Other common symptoms include speech impediments, impaired physical development [2], and increased sensory sensitivity [3], making these assumed processes our design would need to help the user carry out. Finally, the need of the user to move around, feel comfortable, and be safe meant that our design would have to be constrained so as to not limit movement, be loud, heavy, or unsafe.

Due to the personal nature of user artifacts, we had to refer more to our specific primary persona, which will be discussed in the Users section (page 2). Our primary user required a g-tube, stander, and wheelchair, so our design had to be compatible with these devices. Overall, this is a category which necessitates more research into the individual user, which we noted in our chart. The variety of artifacts required by a single potential user made it apparent that simplicity and portability would be a major concern of our design so as not to interfere with the user's existing solutions.

Potential environments included the user's home, daycare, and where they receive physical/occupational therapy. We could assume that most users would also likely need to visit the hospital, go to school, and go outdoors. Areas we would have to look more into are environments the user may not encounter now, but would have to in the future, as well as locations they might travel to. Due to the variety and nature of environments considered, our design would be constrained by the types of terrain traversed by the user.

When considering the resources available to us, we referred to the project parameters provided. For example, the given project deadline of December 2 meant that we would have around three months to complete the design. Resources available to us at Waterloo were also certain, such as the 3D printing lab, help from professors and peers, information from the parents' panel, and the Design Equipment Nexus (DEN). One possible resource which would require more research was government funding. However, as such a resource would likely necessitate a lengthy and complicated application process, we deemed it outside the scope of our project and did not pursue it further. Constraints included cost, for us and the user. Space taken up by the design, due to the likelihood of usage in a home, and time needed to set up the device as users such as parents and caretakers have many things they need to do in their day. Overall, due to the constraints on resources available to us, our design would likely require more simplicity when it

came to design and materials used, with an emphasis on accessibility for the target user and use in a home environment.

The full situation of concern chart can be found in Appendix A.

1.2 Users

The primary user of this design will be a toddler between the ages of 2 to 4 with level 4 to 5 (out of 5) cerebral palsy. Based on this information, we first created an empathy map comprising information related to the primary user's needs, current actions, goals and frustrations. This map can be found in Appendix B. Using information from this map, we created the primary persona Taylor, a three-year-old girl with level 4 to 5 cerebral palsy who is currently and will likely remain non-verbal. She requires a g-tube to eat and uses either a wheelchair or stander to move around besides being carried by others. She requires support for her neck and back in order to sit comfortably, as well as constant supervision and help with most daily activities. She possesses good neck mobility, is communicative with her eyes and expression, and has a playful and social personality. Her goals are to increase her ability to interact with other children her age, perform imaginative play independently, develop her fine motor skills, and interact with more physical mediums rather than a screen. Factors which make it more difficult for her to reach these goals include her non-verbalism, pain felt when she engages in physical activity, the screen-based nature of many existing solutions, and her inability to approach others physically by herself.

The secondary user of this design is likely to be a parent, occupational therapist, or other caregiver of the toddler. The secondary persona we created is Sarah, Taylor's mother and primary caretaker. Her goals are to give Taylor as much independence as possible, and to aid Taylor in her development. She is frustrated by the lack of support provided by the government and medical professionals, the amount of time required on a daily basis to support Taylor's growth, and the slow speed of Taylor's progress.

Taylor and Sarah's detailed personas can be seen in Appendix C.

1.3 User Needs

Based on the goals and frustrations of the primary and secondary user, we developed groups of user needs our design had to address.

Primary user need	Secondary user need	Needs group
• Develop fine motor skills	• Not applicable	Encourages freedom of movement
• Maintain safety	• Keep child safe • Encourage independence of child	Safe
• Long-lasting solution	• Minimize cost of buying new solutions	Long-lasting
• Sit comfortably • Neck support • Comfortable for longer periods of usage	• Not applicable	Comfortable
• Design maintains structural integrity • Stay in position intended by solution	• Handle design without fear of breaking it	Stable
	• Intuitive to set up • Minimize time used to set up	Easy to use

Table I: User needs groups

Finally, using our research and work done thus far, we developed the following situation of impact statement:

Design a device to be used by a toddler with level 4-5 cerebral palsy with the help of their caretaker in an aquatic environment to allow for exercises to improve limb mobility and limb strength that is comfortable, safe, and simple.

2. Design Requirements and Constraints

Requirements (in order of importance):

- R1: **Safety** — The neck and back must be supported and the head must always be above water.
- R2: **Buoyancy** — Corresponds with safety, since buoyancy keeps the child's head above water.
- R3: **Comfortability** — It must be comfortable so there is no pain when resting in the device.
- R4: **Adaptability around required equipment** — Must not interfere with a G-tube
- R5: **Promote physical activity** — Requires a large range of motion in the limbs

Constraints (in order of importance):

- C1: **Cost** — Must not exceed the critical value of \$500.
- C2: **Durability** — Must last at least the critical value of 1 year.
- C3: **Weight** — Must be less than the critical value 15kg.
- C4: **Absorbability in water** — Must absorb no more than 4kg.

2.1 ISO Standards Document Based Requirements

The Absorbability in water constraint (C4) is also directly addressed within the ISO 12402 [4], to achieve a realistic benchmark for the density of the device after water absorption, the ISO states “The samples shall then be immersed to a depth of 50 mm in fresh water at room temperature for 24 h. The initial volume of each sample shall be determined immediately after the water soak specified by measuring and recording the initial water level, in an inclined manometer tube to the nearest millimetre.” [5]. This provides an accurate benchmark to determine the exact amount of water that the system absorbed. Using this test, we can accurately address the density change to mitigate the weight change of the system. Furthermore, the ISO 12402 standard addresses the durability (C2) of the product. To meet the required standard for the device strength, “each sample shall support for 5 min, without breaking, disengagement, or similar condition” [5]. This test will be used to meet the standard for durability with PFDs and will allow us to calculate the tensile strength of the device.

2.2 Ethics Based Requirement

The ethics-based requirement included is the adaptability around required equipment (R4). This embodies key ethical principles from the PEO Code of Ethics, the “Ethical by Design – A Manifesto” paper by placing a high priority on complementing “differing needs, abilities, viewpoints and morals,” focusing on the needs and abilities [6]. This requirement highlights the importance of making the device usable for all children with cerebral palsy, accommodating for their different kinds of preexisting medical devices (like a G-tube), or other constraints they may have. Based on a secondary stakeholder, a mother with a child who has cerebral palsy, many devices for toddlers, whether it be specifically for children with disabilities or just general products, she often must make slight changes and modifications to accommodate her child’s differing abilities. By prioritizing this ethics-based requirement, we can enhance the overall user experience, creating a device that is trusted and is able to be used by a wide range of children with cerebral palsy.

3. Engineering Specifications and QFD

3.1 Needs and Engineering Requirements

The main stakeholders that our QFD (refer to Appendix E) accounts for are the user of the product and the caretakers of the user. For the primary user, we identified that their main need was to remain safe, which is especially important due to our problem space in water. Additionally, the design must also accommodate for equipment that individuals with CP may require, be comfortable to use, and per our design project purpose, it must help the user engage in physical activity. One thing to note is that promoting physical activity is the primary goal for the secondary user.

Due to our concern for safety, the engineering requirements we identified correlate mainly relate to it, including, the distance of the user's head above water, the durability, the weight of the device, the stability of support materials, and the absorbability in water. After the mentorship session, we realized that we had to not only consider that the head must be above water at all times but that it must be above the waves that may occur from potential disturbances in the water that are consistent with pool conditions. The durability of the design is crucial to maintaining the safety requirements we set even through constant use which correlates to the stability of support materials. Another requirement was that the weight must be reasonable to allow for the buoyancy of the design and to allow caretakers to transport the device with more ease. A consideration we made was that the design must have low absorbability in water to allow for the minimization in an increase in weight which, in turn, could cause the device to be unsafe and less portable. We also considered lower costs to be an important engineering requirement to allow our design to be accessible to those who need it and a simplistic design that keeps setup time low as well as easy to use. Looking ahead, the accumulation of these requirements points us into creating a simple design which ensures the safety of the user as its foremost objective without sacrificing the user's comfort and limb mobility.

With the different considerations mentioned above, we created the different importance values for the What vs. Who section of the QFD. As a group, we decided that physical safety and device safety were both one of the primary needs for both groups of users, so we agreed to give them an importance of 25. Buoyancy and User Comfort were given 15 each for the primary user but only 5 for the secondary user as it was less of a need for them. Through meeting with a mother of a child with CP, we felt that 30 was a reasonable importance number for physical activity in the secondary user column as she mentioned that one of her top goals was to help her child engage in more physical activity. For the same need but for the primary user instead and the need for a device that is adaptable around required equipment, we only set it as 10 as both functions are important but compared to everything else, they are less important. The reason we weighed adaptable around required equipment more than user comfort for the secondary user is that if the device is not adaptable, the device will not be able to be used as intended and may then cause a risk to the primary user's safety. Additionally, some modifications may have to be implemented onto the device by the secondary user which may decrease the effectiveness of it whereas the device may be uncomfortable, but it can still meet all the other user needs including promoting physical activity.

3.2 Engineering requirement Synergies and Tradeoffs

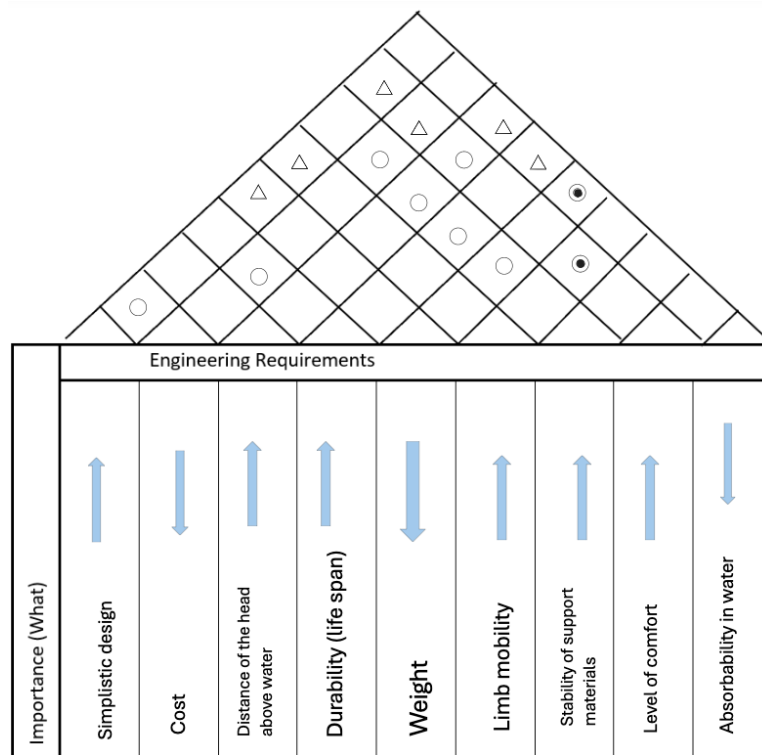


Figure 1: The engineering requirements synergies and the interdependencies. The triangle is a connection of 1, the hollow circle is a connection of 3, and the circle with a concentric inner circle is a connection of 9.

Understanding interdependencies are a key aspect to developing an effective design that best suits the given engineering requirements. Through the development of our QFD, we analyzed the interdependencies between our requirements and recognized that the relationships between “Limb mobility” and “Level of comfort” as well as the relationship between “Weight” and “Absorbability in water” to have the highest correlation.

When developing a comfortable option that is intended for sensitive individuals to wear, we realized that restricting limbs with an overly complicated design would significantly reduce the level of comfort children would experience [7]. Our design now allows free mobility of arms and legs to help accommodate the needs of the primary stakeholder.

To create a design that is able to easily float in a body of water, it is crucial that unnecessary mass is not added to the system. To help counteract this, it is crucial that our device does not absorb excessive amounts of water. This interdependency allowed us to realize the importance of non-absorbing materials to create a lighter prototype. This would also allow easier transportation of the device once it is done being used in the water, providing a more enjoyable experience for the secondary stakeholder.

3.3 Benchmarking and Metrics

For the current solutions, we assessed it based on our vision of an ideal device. For physical safety, we envisioned a clear support system to keep the neck secured in place and in a position where the spine is fully supported. Device safety and buoyancy were assessed based on checking the safety guidelines and the possible safety risks each presented. Through the research we did, both the PFD-A and the Mambo Baby Float were very safe to use but the Otteroo LUMI, due to being an inflatable float, had some major safety concerns. They had a voluntary recall in 2015 for the durability of the floats [8]. Even though Otteroo claims to be perfectly safe [9], we found a warning

issued by the FDA on baby neck floats due to the risk of injury with the strain it puts around the neck or even death through drowning or suffocation [10]. This is a major concern where safety is the primary concern for our design and is especially important because of our problem space with a user with high level cerebral palsy. Along with the Otteroo website reiterating the importance for the check of leaks prior to usage [11], we decided that our design will not incorporate the usage of inflatables due to the hazard of a puncture occurring or a durability issue. Foam as the main material to provide buoyancy is something we have already taken into consideration as seen later in Design 1. The fourth user need was adaptability around required equipment, which, based on our persona, is a g-tube. The Otteroo LUMI does not get in the way of the g-tube as it focuses on only the neck region, but the Mambo Baby Float and the PFD-A are designed with several straps around the midsection, which offers comfort and lots of adjustability, but it may interfere with the g-tube. The major downside to the Otteroo LUMI is that, because it focuses on the neck, it puts a lot of strain on that region alone. Even if it does not harm the user, there are several reviews by parents on the neck float mentioning the tightness of the neck float around the child as the size did not fit them properly [12]. Lastly, none of our competitors do a particularly good job promoting physical activity. The Otteroo LUMI and the PFD-A tie at a 3 for a good range of motion but both are still constricted in their respective ways. The Otteroo LUMI restricts the movement of the head and limits the range of motion of the arms to below the floatie and the PFD-A restricts user to a supine position. The Mambo Baby Float restricts the movement in the user's arms as the arms are limited to always moving above the float or on the float.

Considering the different weaknesses and strengths of the current solutions, the competitor that stood out to us was the PFD-A, a life jacket specifically designed for those with special needs and reduced mobility [13]. It gave us good targets for the safety metrics of our design. This included the stability of support materials, the level of comfort, and limb mobility allowed. They also come in multiple sizes to accommodate a full range of body sizes and weights. The only requirement that the PFD-A severely

		Now		
		Otteroo LUMI (Baby Neck Float Ring)	Mambo Baby Float	Life Jacket (PFD-A Small Size)
User Needs/Functions		1	4	5
Safety	Physical Safety	2	4	5
	Device Safety	2	4	5
	Buoyancy	5	3	3
Comfort	Adaptable Around Required Equipment	2	4	4
	User comfort	3	1	3
Physical Activity	Promotes Physical Activity	3	1	3

Figure 2: “Now vs. What” section of QFD

underperforms is the cost aspect. The cost for a single PFD-A is \$190.00 CAD for the size suitable for the age group of two to four [13]. We want to create a less expensive device that offers the same safety, support, and comfort with a little more freedom in mobility. Cost was one of the easier targets to set as we based it off of the current solutions in addition to considerations of what the secondary persona would have wanted. We believed, as a group, our device should be accessible to everyone who needs and wants to use it. This meant lowering the barrier that the cost may create. In the end, we settled on the cost of \$100 CAD as our target. For the Likert scale metrics, we placed our minimum threshold at 5 which we perceived as a barely satisfactory rating. As our target, however, we wanted a 7 to show an overall good to high user satisfaction. From the aforementioned mentorship session (3.1 Needs and Engineering Requirements) we decided to take inspiration from our competitors to determine the value of our target for the distance of the head above water. Unfortunately, we could only find this value for one of our competitors, so we solely based it on the height of the Mambo Baby Float. We rounded up to 15cm because our specific age range is larger than the float which has an age range of 3 to 48 months [14]. The warranty for all three competitors lasted only one year so we assumed that the life span of their devices was all one year. We wanted to challenge ourselves by creating a device that could surpass the others in durability. This would hold our device to higher safety standards, but it will save our secondary persona money and peace of mind. The weight target of 5kg for the device itself was an estimate on what we felt was light enough to carry around conveniently and to allow for the floatation of the user. The weight we wanted to support with the device was 25kg which we came up with when we compared the average weight of 4-year-old children [15]. For the absorbability, the change in kg we chose was 1 kg for the convenience of the secondary user who has to take the device out of the water after use.

4. Functions & Means

Using the design tools such as the QFD, SOC, and the function-means tree, we were able to identify six needs for our water assistive device. Each need is addressed by a few functions of our device which focusses on the usability, safety, and comfort to meet the specific requirements of the stakeholders. Visual aids from the function-means tree are detailed in appendix E.

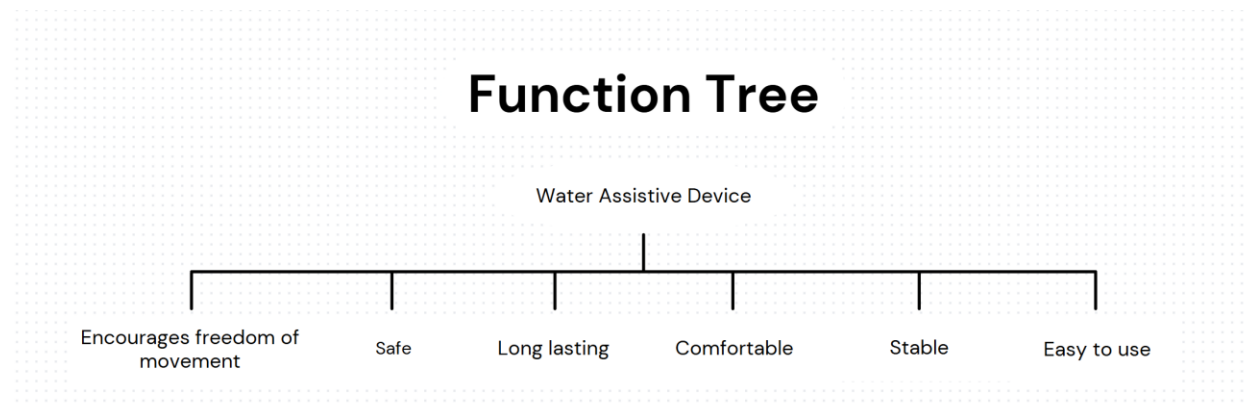


Figure 3: This function tree highlights the essential functions relating to the water assistive devices.

1. Encourages Freedom of Movement

To address this need, the device must have minimal constraints, specifically around the limbs. The only constraints provided must be the neck, back and upright support. This function also addresses the requirement of promoting physical activity since it encourages free movement. It is essential for all children to receive some sort of physical activity, even if they have physical disabilities, this function of the device would allow them to experience this. The way that this function will be implemented in the device is through the main torso constraint which keeps the user upright, while allowing them to freely move their arms and legs in the water. This is done through attachments on the top and bottom part of the torso constraint to the sides and back of the floatation part of the device. Materials were selected for flexibility without compromising support.

2. Safety

Safety is the most important need to be addressed in our assistive device since it was ranked the highest importance in the QFD. This aligns with the ethical obligations and addresses critical constraints such as preventing injury, limiting excessive movement, and ensuring the device is fail-safe. To address this important need, the device must have certain level of buoyancy such that the child's chin is a certain distance above the water to prevent drowning. The device must also have padded support, waterproof materials and secure fastening mechanisms like Velcro and quick-release buckles so that the child is always supported in the device, with proper neck and back support.

3. Long-lasting

The durability of the device is a crucial need for the users, especially for the secondary stakeholders (the caregivers), since the longer this device can last, the less money they must spend on repairing, or purchasing a new version of the product. To address this need, the device must withstand prolonged exposure to water, wear and various environmental factors. Durability ensures that the device is consistently safe to use over long periods of time. We chose materials that are waterproof so that they are resilient to wear, ensuring it remains functional even with regular use in water.

4. Comfortable

Comfortability is essential for the primary user of this device since it is encouraging frequent use of the device aligning with user needs highlighted in the QFD and personas. The function to address this need is that the device should provide a comfortable fit and accommodate for a g-tube and other necessary personal constraints. To implement this, the device incorporates soft padding over device, especially over edges, in case of sudden movements. Adjustable straps also allow customization to fit different body types and sizes.

5. Stable

In order to address this need, the device must maintain a steady and balanced position for the user, minimizing tilting and tipping. This is important because it ensures that users remain safely positioned without risk of overturning, which addresses key safety and support needs in the Situation of Concern and QFD. Stability also contributes to user confidence, making the device less intimidating for users with cerebral palsy. This function will be implemented through strategic placement of buoyant materials around the torso constraint and low center of gravity positioning. This would allow a stable, upright position for the child, preventing any tipping from external factors or sudden movements.

6. Easy to Use

This need will simplify the setup, fitting, and usage to minimize the time and effort required from the secondary stakeholders (the caregivers). This easy-to-use device will reduce stress and time to set up which is crucial because it leaves less room for error when using. This need was noted in the SOC and the QFD as a top priority for caregivers. To implement this function, the device will have a simplistic design, features such as quick-release buckles and Velcro will be used as well as light materials which makes it easy to transport.

Each function and means addressed in this section have been carefully created to meet the needs of stakeholders, ensuring that the device is safe, comfortable, and effective in all situations. By incorporating the QFD and SOC analysis, these functions fulfill the device's purpose. The appendix includes the function-means tree and the morphological chart to visualize how each feature contributes to the overall design.

5. Design concepts

The following ideas that were sketched were developed by the group in a process similar to the diverging - converging ideation cycle. Everyone brainstormed possible solutions and shared their ideas freely with the group. Once everyone had shared their ideas, the concepts that the group liked were then elaborated upon and discussed further. This was sometimes done by creating quick, rough sketches. This narrowed down the number of ideas, as some ideas were deemed unrealistic or were rejected during this phase. After narrowing the concepts down to three, more details were added as the final sketches were made.

5.1 Design 1- Flotation Device

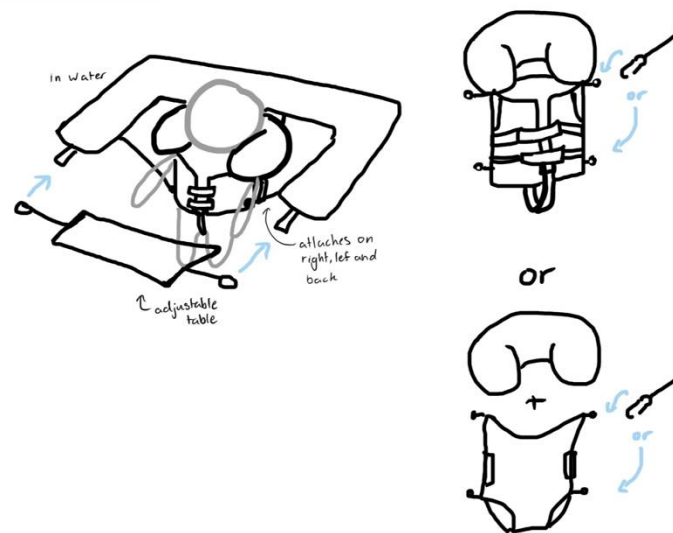


Figure 4: Sketch 1 of Flotation Device

Design 1 is a flotation device similar to a lifejacket, but specifically designed to support a toddler with cerebral palsy (CP). This design features three main structures: The neck support, body support and outer ring. They will be made from durable, water-friendly materials, such as waterproof foam. The neck support is extremely important as it ensures that the child's head is always above water and supports their neck. Safety is a primary concern that was considered throughout the ideation and creation of this design. The body support is what secures the toddler in place. It is important that it is adjustable to ensure the users comfort, a user need outlined in the function tree and QFD. There are two variations of the body support. One (top right) where the child is primarily supported by their armpits, like a lifejacket, and two (bottom right), where the child is supported from underneath, like a harness. Both attach to the outer ring via clips and a cord. The outer ring keeps the child afloat and keeps them from rolling over. Should the child start to roll over, the cord attaching from their body support to the outer ring stops them. These aspects of the design assure that the child is safe, afloat and supported. Another key aspect of this design is that it leaves the toddler's limb completely free to move under the water. Most of the equipment is on top of the water, nothing is around the limbs, except for the body support which can be adjusted as necessary. This encourages physical activity, a priority in our situation of concern.

Many advantages to this design come from its padded, waterproof material, such as waterproof foam. Softer materials protect the toddler from hitting any hard surfaces and are also durable in water and will likely work for a long time [16]. This means that this design could ideally be used over the span of a few years. Another advantage is that this device can be used in any water depth without adjustment because it is buoyant. However, the buoyancy of this design has weaknesses as well. There is nothing anchoring the toddler to a specific place, which leaves them liable to drift off. This is one of the reasons this design would not be compatible with open water, along with the fact that waves could splash water into the child's airway. Another drawback of this design is the limited view. The neck support, while essential for stability, may limit peripheral vision.

5.2 Design 2- Pool Support Device



Figure 5: Sketch 2 of Pool Support Device

Sketch 2 depicts another design that supports a toddler with CP in the water. The device is similar, and the body and neck support would have the same structure and function as the ones outlined in the description of Sketch 1. However, contrary to our first sketch, this device is non-buoyant and sits on the bottom of the pool to support the child. The device features a base that can be adjusted for different pool depths, with wheels that can be locked, allowing it to either stay in place or move as desired. The upper section of the base will be padded with a table on the top. The body support then attaches to the base with a mechanism that fixes it to the base. This ensures the child is secure, unable to roll their face near the water. Though the child is secure, their limbs are free to move. Since the base must be durable in water, it will be made from waterproof material, such as plastic.

The strengths of design 2 are provided by the strong base. The base is that it is padded to ensure that there are no sharp or hard edges that may pose a danger. It ensures that the child will not drift off and keeps them stable. Stability is a function outlined in the function tree and is critical for safety. However, this also means that the child is unable to move independently in water. Though the base is immovable for safety, this may negatively impact user experience. Another disadvantage is that this design takes up a lot of space and needs to be heavier to remain stable. This can make transportation of this device difficult. Another disadvantage to this design is that adjusting for different water levels could be tedious.

5.3 Design 3 - Adaptive Flippers

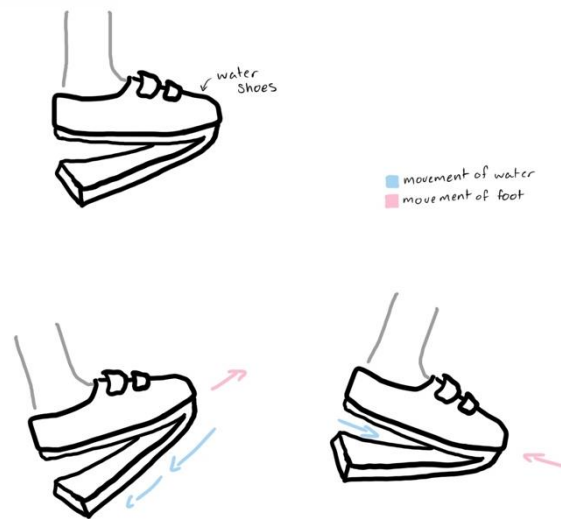


Figure 6: Sketch 3 of Adaptive Flippers

This third design shows an idea for flippers that facilitate water movement for with CP. They are intended to be paired with a floatation device that safely supports the toddlers head above the water, such as design 1. Unlike regular flippers, that require a specific kicking movement to push the user forward, these flippers capture water when the users kick is uncontrolled or more like bicycle kick. While normally, kicking uncontrollably would not move the child anywhere, with these flippers they will be able to move around the aquatic environment. This will encourage physical activity, especially in the legs. The flippers would be tailored to the toddler's foot size, with Velcro straps to ensure the user's comfort. We would ensure that the flippers are lightweight and made from slightly flexible waterproof material such as rubber or plastic.

This design is different from the others as it requires existing equipment to keep the toddler safe in the water. The user needs and engineering requirements are not fulfilled without that accompanying support. Furthermore, these flippers will need to be replaced as the child grows, though the fit is adjustable due to the Velcro straps, the length is not. This design's strengths stem from its simplistic design, a desire mentioned in the situation of concern. These flippers are small, compact and easy to use. This makes them easily portable and usable in any safe aquatic environment. Also, the flippers do not require relatively expensive material or a lot of material, which would greatly minimize cost, which is a constraint.

6. Gallery Walk and Mentorship Reflection

The gallery walk was an extremely valuable experience to receive feedback from peers on our various design concepts. During the engineering design process, it is easy to narrow your perspective and disregard the potential flaws associated with each design. An outside view from a wide variety of individuals provided us with feedback that we are able to integrate to further refine our concepts.

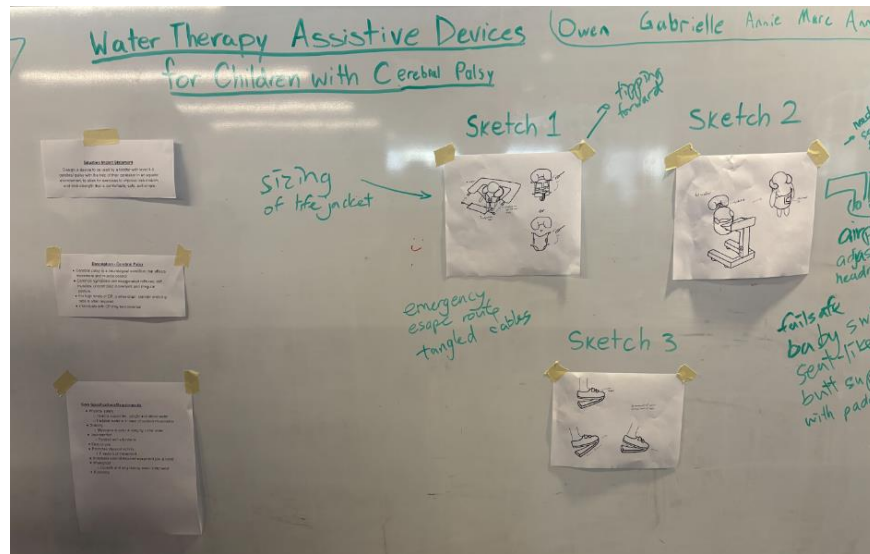


Figure 7: The board utilized during the gallery walk which showcases design sketches, Situation Impact Statement, description of problem space, and core specifications.

During the gallery walk, we separated our feedback into three distinct sections:

1. Positive comments: Parts of concept that were appreciated and should be focused.
2. Areas of concern: Parts of concept that are flawed and contradict our user needs, (Figure (QFD)).
3. Areas of improvement: Parts of concept that need further development to support primary stakeholders.

Through this, we were able to effectively organize and integrate the feedback for our concepts, most notably in issues regarding our device stability, the jacket sizing system, and which materials would be suitable for sensitive individuals.

The mentorship session also helped further our understanding of the timeline regarding the engineering design process. They helped inform us about the difficulties regarding prototyping and what precautions should be taken before our prototypes. More specifically involving the physics calculations involving the buoyancy required to support children within a specific weight interval. They also provided valuable criticism to the vagueness of our problem space, suggesting that we narrow it to a specific type of cerebral palsy and a specific body of water where our concepts will be implemented.

Overall, the gallery walk and mentorship session provided us with unique feedback in various areas of the engineering design process to help us develop and refine our design concepts.

7. Conclusion

The engineering design process is a lengthy and complex process which requires lots of meticulous research to develop a realistic and feasible solution. We started by developing a realistic needs assessment based on the needs of a stakeholder interviewed during a panel discussion. This fostered our passion in the space and helped us understand the current systems that fail support the needs of those who require them most. We aim to create a therapeutic assistive device in an aquatic environment for children with level 4-5 Cerebral Palsy. Through the development of realistic personas, we were able to better understand the complexities of the issues within the day to day lives of the primary and secondary stakeholders. We then utilized the fundamentals of a QFD to understand the relationship between the engineering requirements and user needs, as well as to better compare our solution to that of competitors working in a similar problem space. This helped us realize the flaws that competitors designs hold so we can create a more effective solution. Through the creation of various concept sketches, we were able to better envision our solutions and understand how we can incorporate realistic solutions to the issues we realized during the creation of the needs assessment.

Our next steps are to select a design out of the three that we came up with and eventually create a medium-fidelity prototype of that design. We are also planning to update our design history file and logbook to add more detail to the existing elements, as well as summarize the work we have done recently. Currently, we are going over how to perform engineering analysis in class. Once we have a clear idea of how to go about this, we can apply such analysis to our design. Hopefully, this will help us be critical about what parts of our design will work, which will be useful for creating our prototype.

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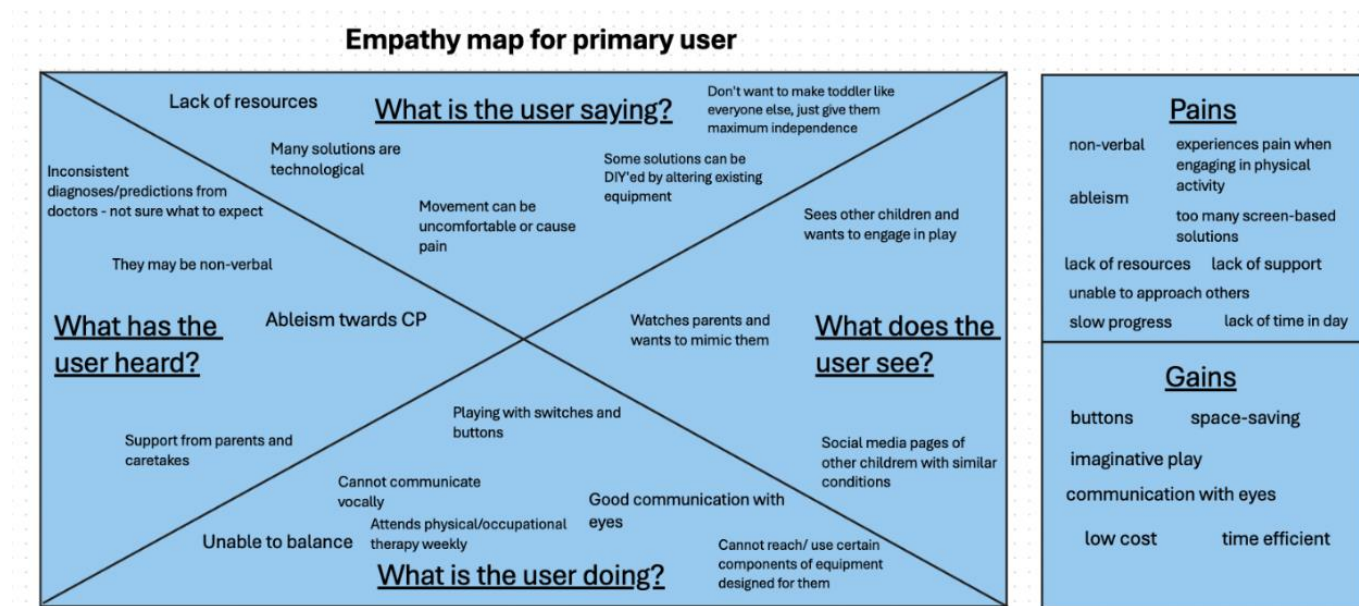
9. Appendix

Appendix A

Situation of concern chart

Cerebral Palsy					
	Stakeholders Who is impacted?	Processes (Jobs, tasks, etc) What do they need to do?	Artifacts (Products, equipment, data) What do they need to use?	Environments Where do they do the tasks?	Resources (time, money, etc) What is available?
Facts What do you know?	- Child, primary stakeholder (Etta) - Parent(s)/Caretaker	- Physical movements (ie. fine motor skills)	- g-tube - wheelchair - stander **Varies per person	- Daycare - House - Therapy places	Resources we have: - time (until December 2) - 3D printing lab - professors, peers, mentors, etc. - Internet/online forums - Information from parents (parent perspective) The Den
Assumptions What do you think you know?	- Physical /OccupationalTherapist - Doctors (pediatricians) - Support systems (other families/children with similar disabilities)	- Socialization (movement or speech impediments) - Developmental activities	- Communicative devices (buttons) -	- Hospital - School - Outdoors (ie parks)	- A small budget - Testing area - Parent input
Required Research What you don't know?	Other support group interactions (Teachers, baby sitters)	- Levels of CP - Limitations of movement caused by equipment such as : g-tube, feeding tube, cochlear implants	- Pre-existing devices	- Future environments - Travel	- Government Grants - Coverages
Critical constraints What it must do or not do	Help people in need, Be accessible to all users, Easy to understand and maintain for users	- Must not Limit range of movement - Must not be loud - Must not be heavy - Must be safe	- Safety concerns - Complexity - Portability	- Terrain - Accesibility of resources	- Cost - Space - Partent/Caretaker Time - Team time

Appendix B

Empathy map

Appendix C

Primary Persona (note: images included were generated using Adobe Firefly)

Primary Persona



Name: Taylor Dubois

Info:

Age: 3 years old
Location: Waterloo, ON
Diagnosis: Cerebral Palsy
Siblings: Brother (5)
Hobbies: Playing with toys (switches and buttons),
Reading books

Personality + Skills:

- Decent neck mobility
- Communicative with eyes and expression
- Social and like to engage
- Likes switches & buttons
- Likes mirroring parents/peers
- Overall very playful

Story:

- Birth complications resulted in cerebral palsy
- Likely to be non-verbal
- 4-5 on the cp scale
- Uses a g-tube, wheelchair and stander
- Attends physical and occupational therapy every week
- Constant supervision and help with everyday activities
- Support for movement and sitting

Goals:

- Increased ability to interact with others her age
- Imaginative play
- Increased fine motor skills and range of mobility
- Interacting with more physical mediums rather than a screen

Needs:

- Time efficient
- Doesn't take up a lot of space
- Low cost

Frustrations:

- Non-verbal
- Feels pain when engaging in physical activity
- Many solutions are very screen based
- Lack of support and resources
- Ableism
- Unable to approach others physically

Influences:

- Parents
- Siblings
- Therapists and doctors
- Daycare classmates and teachers
- Extended family

Secondary Persona (note: images included were generated using Adobe Firefly)

Secondary Persona



Name: Sarah Dubois

Info:

Age: 41 years old

Location: Waterloo, ON

Hobbies: Reading, baking,
building

Occupation: N/A

Primary caretaker of Taylor

Personality + Skills:

- Outgoing, positive
- Hopeful, smart

Story:

- Middle age Mom, 41
- Healthy, no past medical history
- middle class
- Married to John Dubois

Goals:

- To give child as much
independence as possible
- Raise Taylor

Needs:

- Time efficient
- Doesn't take up a lot of
space
- Low cost

Frustrations:

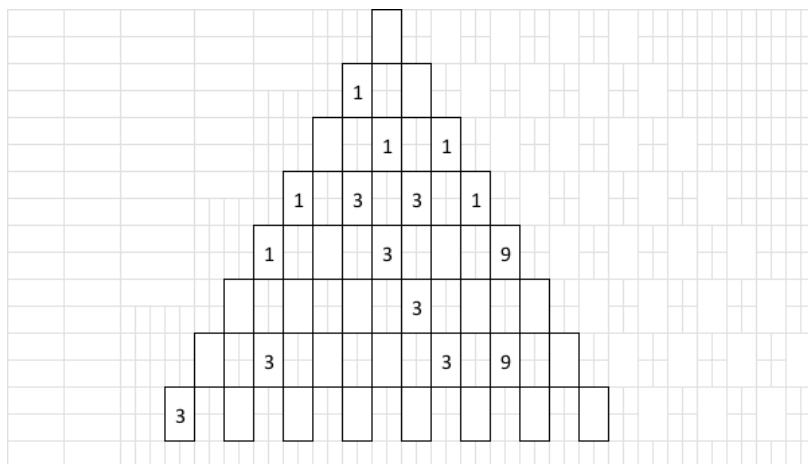
- Lack of support (from others and
from government)
- No time
- Ableism
- Slow progress
- No cooperation from doctor
professionals

Influences:

- Parents
- Husband
- Friends
- Others who have
children with similar
disabilities.

Appendix E

Quality Function Deployment (QFD) Chart



User Needs/Functions		Who		Engineering Requirements									Now		
		Primary User (Child with CP)	Secondary User (Caretaker)	Simplistic Design →	Cost ↓	Distance of head above water →	Durability (life span) →	Weight ↓	Limb mobility ↓	Stability of support materials ↑	Level of comfort →	Absorbability in water ←	Otteroo LUMI (Baby Neck Float Ring)	Mambo Baby Float	Life Jacket (PFD-A Small Size)
Safety	Physical Safety	25	25	1	1		3			9	3		1	4	5
	Device Safety	25	25		1	9	3	3		1		9	2	4	5
	Buoyancy	15	5		1	9	1	3				9	2	4	5
Comfort	Adaptable Around Required Equipment	10	10	3	1						9		5	3	3
	User comfort	15	5		1		1	1	3	3	9		2	4	4
Physical Activity	Promotes Physical Activity	10	30					1	9	1	3		3	1	3
		Units iJkert scal CAD (\$) cm Years kgs iJkert Scal Kgs iJkert Scal Δkgs													

Importance (How important are these Hows to Who?)															
Primary User (Child with CP)		55	90	360	180	145	135	305	330	360					
Primary User (Child with CP)%		3%	5%	18%	9%	7%	7%	16%	17%	18%					
Secondary User (Caretaker)		55	70	270	160	125	285	45	300	270					
Secondary User (Caretaker)%		3%	4%	17%	10%	8%	18%	3%	19%	17%					

Competition (Now vs. How by How Much)															
Otteroo LUMI (Baby Neck Float Ring)		9	60	N.D.	1	N.D.	8	16	6	N.D.					
Mambo Baby Float		7	180	14	1	5.9	N.D.	18	9	N.D.					
Life Jacket (PFD-A Small Size)		6	190	N.D.	1	N.D.	9	18	8	N.D.					
How much is good enough for each How?															
Target (Delighted)		7	100	15	3	5	7	25	7	1					
Threshold (Disgusted)		5	150	10	1	10	5	20	5	4					

Sample calculations of Importance on the QFD:

$$Importance = \sum (Weight \times correlation)$$

$$Importance \text{ of engineering requirements for the primary user} = (25 \times 1) + (10 \times 3) = 55$$

$$Importance \% = Importance / Total Importance$$

$$Importance \% \text{ of engineering requirements for the primary user} = 55/1960 = 3\%$$

Appendix F

Function-Means Tree

